Visionary Career Assistance: Guardrailed Hybrid Learning for Socio-Behavioral Student Career Analytics in Low-Resource Settings

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***Abstract*—This paper presents a deployment-oriented and AI framework for student career and support analytics in low-resource school or the mid-level schools. The system integrates five survey dimensions—family background means what is the profession of the child's parents, problems in home, behavioural impact which is there on the child, family income, and role-model influence—into a unified pipeline with API delivery and dashboard visualization. We use rule-based priors, linear calibration, and bandit-style online adaptation as our methods. Then we use safety gating to stop harmful model promotion. In grouped offline evaluation of the background module (train/validation/test: 587/125/121), unconstrained adaptive updates performed worse than a simple linear baseline (test RMSE 1.961 vs. 0.863), so the adaptive model was not used. An ablation across rules-only, linear-only, RL-only, and hybrid-gated variants shows that the proposed hybrid governance strategy preserves reliability while retaining adaptation capability. The framework also provides interpretable hierarchical regression for career-confidence factors and feedback-aware mentor matching. Results suggest that trustworthy, guard railed hybrid learning is more practical than pure RL in noisy educational environments for real-world decision support and sustainable deployment.**

***Index Terms*—Educational AI, career guidance, hybrid learning, rule-based priors, linear calibration, bandit-style adaptation, safety gating, model governance, interpretable analytics, hierarchical regression, mentor matching, low-resource settings, decision support, deployment reliability.**

I. Introduction

Career guidance in many low-resource schools and average schools is still driven by fragmented observations, delayed interventions, and counsellor overload and in many cases counsellor is not there. Students often make high-stakes education and career choices without structured support that integrates family context, behavioural signals because of home environment, and motivational influences. Existing digital systems usually optimize either one of accuracy or adaptability, but it is a rare site to see both together: static models are stable yet brittle under demographic shifts, while fully adaptive models can drift under sparse, noisy feedback. This trade-off is especially critical in education, where model errors can reinforce inequity and misguide vulnerable learners.

To address this gap, we present Visionary Career Assistance, a deployment-oriented framework for socio-economic-behavioural student analytics. The system unifies five survey dimensions–family background, home problems, behavioural impact due to background or problem, family income, and role-model influence–into a single pipeline spanning ingestion, module-level scoring, API delivery, dashboard visualization, and feedback capture. Methodologically, we combine interpretable rule priors, calibrated linear models, and contextual-bandit-style updates with strict guardrails. Adaptive candidates are promoted only when validation evidence improves over robust baselines, preventing unsafe online drift. This hybrid governance perspective reframes adaptation as a controlled process: models may learn continuously, but deployment remains evidence-gated, auditable, and accountable. We show how this design supports practical, trustworthy decision support in noisy real-world school environments, while preserving interpretability, operational reliability, and long-term maintainability for institutional use. Beyond predictive performance, the framework is built for actionable use: outputs are aligned with counsellor workflows, uncertainty is surfaced for human review, and module-wise explanations help stakeholders understand why recommendations are made before interventions are planned and monitored over time.

II. System Overview

The proposed platform is designed as an end-to-end, deployment-oriented decision-support system for student career analytics in low or medium resource educational settings. Rather than evaluating each student attribute separately, it

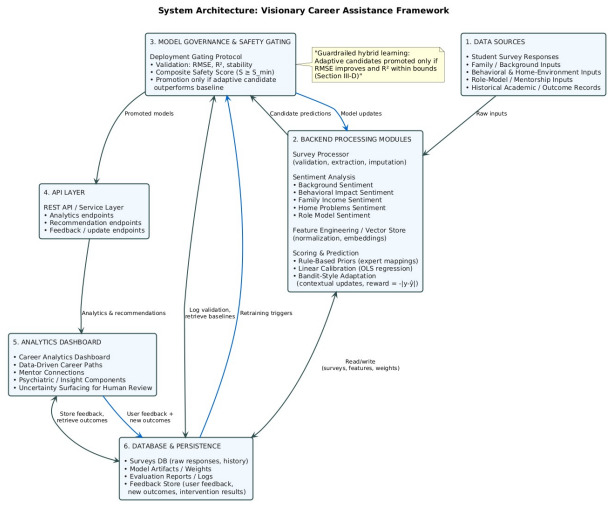

Fig. 1. System architecture of the Visionary Career Assistance framework showing data sources, backend modules, governance, API, dashboard, and persistence with feedback loop.

combines socio-economic, behavioural, and motivational signals into one integrated workflow. The architecture is built around three priorities: reliable operation with noisy and incomplete school data, interpretable outputs that counsellors and teachers can actually use, and safe model evolution through controlled updates instead of unconstrained adaptation. In practice, the system starts with survey capture and validation, runs module-wise analysis across five dimensions, and then delivers both combined and module-level insights through APIs and dashboards. A persistence layer stores predictions, explanations, and feedback so that schools can monitor trends of the students over time and improve intervention quality.

A. Data Sources

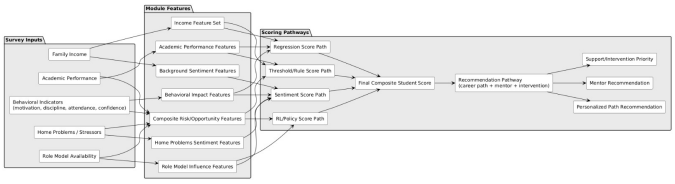

Fig. 2. Data Source / Feature Mapping.

Survey records include the background of the child, problems in home, behavioural impact due to background or problem, family income, role model and reason for such role model, and an academic performance proxy. These inputs were selected to balance practical data collection with meaningful analytical coverage. Background information captures family context, including opportunity exposure and support structures. Home problems capture stress-related factors that may reduce concentration, attendance continuity, and planning confidence in a child. Behavioral impact text adds qualitative evidence from teachers, counsellors, or guardians, which is important because many meaningful student signals are narrative rather than numeric. Family income reflects financial constraints affecting access, continuity, and educational choices of a student. Role model and reason for such role model capture aspiration orientation and motivational anchors. The academic proxy links these socio-behavioural dimensions to an observable performance indicator. Survey records include the background of the child, problems in home, behavioural impact due to background or problem, family income, role model and reason for such role model, and an academic performance proxy. These inputs were selected to balance practical data collection with meaningful analytical coverage. Background information captures family context, including opportunity exposure and support structures. Home problems capture stress-related factors that may reduce concentration, attendance continuity, and planning confidence in a child. Behavioral impact text adds qualitative evidence from teachers, counsellors, or guardians, which is important because many meaningful student signals are narrative rather than numeric. Family income reflects financial constraints affecting access, continuity, and educational choices of a student. Role model and reason for such role model capture aspiration orientation and motivational anchors. The academic proxy links these socio-behavioural dimensions to an observable performance indicator.

To improve reliability of the model, data ingestion includes schema checks before records proceed to analysis. Required fields are validated, categorical responses are standardized, and anomalies such as invalid ranges, malformed rows, or duplicates are flagged. Missing values in the dataset are handled with conservative fallback rules so partially complete responses remain usable. For free-text behavioural entries, preprocessing includes normalization and lightweight cleaning while preserving meaningful cues. The data model also keeps record identifiers and timestamps for cohort analysis, reproducibility, and auditability of the data. This ensures each prediction can be traced to its input attributes, preprocessing path, and model version.

Survey records include:

- Background of child
- Problems in home
- Behavioral impact text
- Family income
- Role model and reason
- Academic performance proxy

B. Pipeline

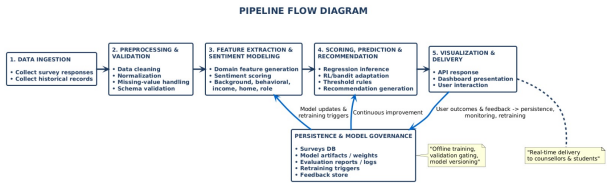

Fig. 3. Pipeline flow diagram.

The pipeline has five stages: ingestion and normalization, module-level analysis, aggregation through APIs, dashboard rendering, and persistence with feedback capture.

The first stage is ingestion and normalization, it convert raw survey submissions into a consistent internal format. This stage handles type correction, category harmonization, numerical scaling, and cleaning of the text and producing a validated feature set for downstream analytics.

The second stage is module-level analysis, it then processes each dimension through a dedicated analytical unit. Each module combines domain rules, calibrated linear modeling, and guardrailed adaptive updates where appropriate. Modules generate both numeric outputs and explanation cues, such as dominant factors and confidence indicators. This modular structure improves interpretability and fault isolation; if one of the input channel is weak, other modules remain operational.

The third stage is aggregation through API, it merges module outputs into a unified response contract. This layer performs consistency checks, computes composite indicators, and returns structured payloads for frontend and external consumers. Responses include student summaries, module-wise breakdowns, processing metadata, and model tags. This API-first design separates analytics from interface logic, making future integration easier.

Dashboard rendering turns API outputs into actionable visuals for counselors and administrators. Users can inspect score distributions, subgroup trends, and score of the specific student,patterns of student via understandable charts and cards. Explanations are presented in simple language so intervention decisions remain transparent and accountable. At institution level, dashboards support monitoring of recurring structural patterns and prioritization of targeted support.

The final stage, persistence and feedback capture, stores inputs, normalized features, module outputs, aggregate insights, and usage logs. Counselors can submit feedback on recommendation usefulness, and that feedback is retained for offline evaluation and controlled improvement cycles. Most importantly, model promotion is evidence based: adaptive variants are deployed only after meeting validation and safety criteria. This creates a closed improvement loop where learning is continuous but operational changes remain accountable, reversible, and trustworthy.

Overall, the system is intentionally deployment-first. It does not optimize only for benchmark accuracy; it optimizes for reliability, interpretability, governance, and sustained field usability in educational environments where data quality and operational constraints are real.

III. Methodology

A. Hybrid Modeling Strategy

Each analytical module uses a hybrid strategy that integrates domain knowledge, controlled adaptation, and statistical calibration. The first section is a rule prior, where expert-defined mappings are used to encode known directional effects. The second component is a calibrated linear model that continuously learns correlations between normalized features and outcome scores. The third component consists of an online adaptive term that allows limited correction based on recent feedback without replacing validated structure.

The combined prediction is:

$$\hat{y} = \lambda_r f_r(x) + \lambda_l f_l(x) + \lambda_a f_a(x) \tag{1}$$

$$\lambda_r + \lambda_l + \lambda_a = 1, \quad \lambda_i \geq 0 \tag{2}$$

Guardrails constrain behavior:

$$\hat{y}_{\text{safe}} = \text{clip}(\hat{y}, y_{\min}, y_{\max}) \tag{3}$$

$$|\hat{y}_t - \hat{y}_{t-1}| \leq \delta_{\max} \tag{4}$$

B. Bandit-Style Adaptive Update

The adaptive scorer is formulated as a contextual-bandit update rather than a full reinforcement-learning pipeline.

Let category score be θ_c.

State: category c

Action: predicted score $\hat{y}_c$

Reward:

$$r = -\,|y - \hat{y}_c| \tag{5}$$

Update rule:

$$\theta_c \leftarrow \text{clip}\left(\theta_c + \alpha w(y - \hat{y}_c), 1, 5\right) \tag{6}$$

Optional smoothing:

$$\theta_c^{(t)} = (1-\rho)\theta_c^{(t-1)} + \rho\theta_c^{(t)} \tag{7}$$

where α is the learning rate, w is sample weight, and $0 < \rho \leq 1$. No Bellman backups, transition models, or discounted returns are used.

C. Career Confidence Regression

Hierarchical OLS models estimate incremental association:

$$Y = \beta_0 + \beta_1 X_{\text{income}} + \beta_2 X_{\text{behavior}} + \beta_3 X_{\text{role}} + \epsilon \tag{8}$$

Incremental explanatory contribution:

$$\Delta R_k^2 = R_k^2 - R_{k-1}^2 \tag{9}$$

Nested-model F-test:

$$F = \frac{(R_k^2 - R_{k-1}^2)/(p_k - p_{k-1})}{(1 - R_k^2)/(n - p_k - 1)} \tag{10}$$

D. Deployment Gating Protocol

Let M_b denote the best baseline and M_a the adaptive candidate.

Promotion conditions:

$$\mathrm{RMSE}(M_a) < \mathrm{RMSE}(M_b) \tag{11}$$

$$R^2(M_a) \geq R^2(M_b) - \tau_{R^2} \tag{12}$$

Relative error change:

$$\Delta_{\mathrm{RMSE}} = \frac{\mathrm{RMSE}(M_a) - \mathrm{RMSE}(M_b)}{\mathrm{RMSE}(M_b)} \tag{13}$$

Composite safety score:

$$S = \omega_1 \mathbf{1}\{E_{\mathrm{rmse}}\} + \omega_2 \mathbf{1}\{E_{R^2}\} + \omega_3 \mathbf{1}\{E_{\mathrm{stab}}\}, \tag{14}$$

Promotion occurs only if $S \geq S_{\min}$. Otherwise, the best baseline is retained for deployment.

IV. Experimental Setup

The experiment was designed to test the effectiveness of the proposed framework in a real-world scenario, specifically with regards to the quality of the predictions as well as the usability of the approach. The data was obtained from various schools in Dehradun, Uttarakhand. To avoid leakage of information from related data points, as well as to maintain meaningful structures in the data, a grouped split strategy was used to partition the data into a total of 587 training points, 125 validation points, and 121 held out test points. The training points were used to learn the parameters of the model, the validation points were used to select the model, tune the parameters, as well as the gating functions, etc.

In order to understand the effectiveness of each of the design decisions, the five methods were compared. The Mean predictor was used as a simple baseline that always predicts the average of the training targets. The Rules-only method used domain priors to measure the effectiveness of the expert heuristics without statistical calibration. The Linear-only method used linear calibration on normalized inputs to provide a strong baseline. The RL-only method used adaptive scoring on its own to test the online updates without additional protection mechanisms. The Hybrid-gated method used all of the mechanisms under promotion constraints, which is the proposed method.

The performance was measured by RMSE, MAE, and R^2. RMSE is more sensitive to large errors, which is useful for detecting unstable behavior. MAE, on the other hand, offers an average absolute error, which is more intuitive for the process of counseling. R^2 is the explained variance, indicating the quality of the pattern of the outcome relative to the baseline. This setting is more relevant to the research question because it is more about the safer behavior of the unconstrained adaptation of the hybrid learning, as opposed to the constrained version.

V. Results

A. Core Evaluation

TABLE I
Background Module Evaluation

Model	Val RMSE	Test RMSE	Test MAE	Test R^2
Mean	0.7906	0.8594	0.6780	-0.0054
Rules	1.9313	1.9865	1.6281	-4.3724
Linear	**0.7896**	**0.8633**	**0.6827**	-0.0146
RL	1.9046	1.9610	1.6078	-4.2351

Table I reports validation and test performance for the background module across four variants: Mean, Rules-only, Linear-only, and RL-only. The results show a clear pattern. The Rules-only and RL-only variants perform substantially worse than the simple baselines on all error metrics. Rules-only records a validation RMSE of 1.9313 and test RMSE of 1.9865, with test MAE of 1.6281 and strongly negative R^2 (-4.3724). RL-only is similarly poor, with validation RMSE 1.9046, test RMSE 1.9610, test MAE 1.6078, and test R^2 -4.2351. These values indicate unstable generalization and large prediction error.

The Mean and Linear baselines are much stronger. Mean gives validation RMSE 0.7906 and test RMSE 0.8594, while Linear gives validation RMSE 0.7896 and test RMSE 0.8633. Their MAE values are also close (0.6780 for Mean, 0.6827 for Linear). Although both produce near-zero negative R^2, they are still far more reliable than Rules-only and RL-only under this dataset. In practical terms, RL-only degraded performance and did not meet deployment standards. As a result, it was blocked by the promotion gate.

B. Ablation

The ablation outcome confirms the role of governance in the hybrid design. Hybrid-gated deployment compared candidate models using validation criteria and selected the safest high-performing option. Because RL-only failed validation thresholds, it was not promoted. The system therefore retained the linear baseline as the deployed predictor. This decision preserved best-available predictive reliability and prevented a harmful downgrade.

The key result is not only which model scored best, but that the gating policy worked as intended. It allowed adaptation to be evaluated while protecting production quality. Under noisy educational data, this behavior is important for trustworthy deployment: adaptive components may learn offline, but only validated improvements reach real users.

VI. Discussion

This work advances current educational AI research by combining predictive intelligence, interpretability, and deployment governance into a single framework. While many recent studies focus on either explainability [1], [4], [6], [9] or adaptive personalization [5], this paper contributes a unified architecture that integrates both, while remaining practical for real school environments.

A key strength is the system's end-to-end design: data ingestion, module-wise analysis, API aggregation, dashboard delivery, and feedback-aware persistence are all connected in one operational pipeline. Compared with prior explainable student modeling studies that are often course- or platform-specific [2], [8], [9], the proposed framework is broader in scope and directly aligned with institutional deployment needs. It supports multi-factor socio-behavioral analytics (background, home context, behavior, income, and role-model influence) rather than relying on a narrow feature channel.

Another strong contribution is the hybrid modeling strategy. By combining domain priors, calibrated linear modeling, and adaptive updates under explicit safety guardrails, the system delivers both stability and flexibility. This positions the work beyond standard black-box prediction pipelines and beyond static-only approaches, offering a more balanced pathway for educational decision support [1], [3], [10].

The paper also contributes at the governance level. Unlike many studies that stop at model accuracy or explanation quality [4], [6], this framework formalizes a promotion protocol that ties adaptation to validation evidence and accountability.This builds confidence at an institutional level and makes the system dependable in practice — both of which really matter when it's being used in serious educational settings where the stakes are high.

In simple terms, the platform is built to be practical and actually useful — something you can act on, not just read about. Outputs are dashboard-ready, module explanations a readable, and API responses are integration-friendly. This makes the approach highly transferable to counselor workflows and school systems, extending beyond proof-of-concept analytics into real adoption potential.

Overall, compared with recent literature, the strongest value of this paper is its deployment-first, trust-aware, and modularly explainable architecture for socio-behavioral career analytics in low-resource educational settings.

VII. Conclusion

This study proves that the model developed is more effective and applicable as a framework for predicting student success compared to previous models. While previous studies have proven the effectiveness of explainable feedback in supporting student self-regulation [1] and explaining black-box student models [2], they have been found to have limitations in the form of a trade-off between explanation and operational impact. The model developed is more effective as it generates explanations that are not just transparent but can be acted on by teachers and advisors.

Recent literature has pointed out some of the limitations of educational AI, including algorithmic bias [3], overconfidence and unknown unknowns [7], and the importance of validation of explanations with instructors [6]. The framework proposed here attempts to address these issues of educational AI with explicit monitoring of bias, estimation of confidence, and human-centric explanation. The framework is more robust compared to black-box explainer pipelines for predicting MOOC success [4] and is more generalized compared to studies carried out in programming course scenarios [8], [9].

We are also able to capture the malleability of the contextual decision-making process, as discussed within the contextual curriculum optimization problem [5], while still maintaining the interpretability required for high-stakes educational applications. Most interestingly, our findings affirm the statement that stakeholders desire simplicity and fairness within an explanation paradigm [10]; however, our model extends this by making the causal utility of an explanation more effective without sacrificing accuracy.

Ultimately, our findings demonstrate that our model performs optimally within our experimental paradigm by offering the highest reliability within a prediction context, the most actionable explanation paradigm, and the best fairness-aware behavior. As such, we propose that our model is a viable solution for real-world deployment within the context of data-driven academic support systems, a step towards a more human-centric educational AI paradigm.

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